

7.2.8 Software Problem Resolution Process

NOTE — The Software Problem Resolution Process may also be used or easily adapted to manage, track, and control **software change requests** and other issue categories (e.g., anomalies, defects, non-conformances) throughout the life cycle.

7.2.8.1 Purpose

The purpose of the Software Problem Resolution Process is to **ensure that all discovered problems are identified, analyzed, managed, and controlled to resolution** in a consistent, traceable, and auditable manner.

7.2.8.2 Outcomes

As a result of successful implementation of the Software Problem Resolution Process:

- a) a problem management strategy is developed;
 - b) problems are recorded, identified, and classified;
 - c) problems are analyzed and assessed to identify acceptable solutions;
 - d) problem resolutions are implemented;
 - e) problems are tracked to closure; and
 - f) the status of all problems reported is known.
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7.2.8.3 Activities and Tasks

The project shall implement the following activities in accordance with applicable organizational policies and procedures with respect to the Software Problem Resolution Process.

7.2.8.3.1 Process Implementation

This activity consists of the following tasks:

7.2.8.3.1.1 Establish Problem Resolution Process

A **problem resolution process** shall be established for handling all problems

(including non-conformances) detected in software products, processes, and activities. The process shall:

- a) ensure that each problem is documented, analyzed, resolved, and dispositioned;
 - b) ensure that status is tracked and reported through closure;
 - c) maintain problem records as stipulated in contracts, organizational procedures, or regulatory requirements.
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7.2.8.3.1.2 Categorization and Prioritization Scheme

A scheme shall be established for **categorizing and prioritizing problems**. Each problem shall be classified by category (e.g., defect, requirement issue, documentation error) and by priority (e.g., critical, major, minor) to support **trend analysis, triage, and resolution planning**.

7.2.8.3.1.3 Trend Analysis

Regular **trend analyses** shall be performed to detect recurring patterns or systemic issues in reported problems. Trends shall be documented and fed back into **process improvement, risk management, and QA activities**.

7.2.8.3.1.4 Resolution Evaluation

Problem resolutions and dispositions shall be evaluated to ensure that:

- problems have been fully resolved;
 - adverse trends have been reversed or mitigated;
 - changes have been correctly implemented in the appropriate software products and processes;
 - no new problems have been introduced as a result of resolution actions.
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7.2.8.3.2 Problem Resolution

This activity consists of the following tasks:

7.2.8.3.2.1 Problem Reporting

When problems (including non-conformances) are detected in a software product or an activity, a **problem report** shall be prepared to describe each problem. The report shall include:

- unique identifier;
 - description of the problem;
 - context and environment in which it was detected;
 - severity and priority;
 - affected configuration items and processes;
 - references to related requirements, designs, or tests.
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7.2.8.3.2.2 Investigation and Analysis

Each problem shall be **investigated and analyzed** to determine root cause, impact, and appropriate corrective actions. This may involve defect reproduction, traceability analysis, or impact assessments across system elements.

7.2.8.3.2.3 Resolution Implementation

Approved resolution actions shall be implemented, verified, and recorded. All related **configuration items, documents, and test artifacts** shall be updated as needed. Resolution may include corrective, preventive, or workaround actions.

7.2.8.3.2.4 Closure and Tracking

Each problem shall be tracked to closure through a **closed-loop process**, from detection through investigation, resolution, verification, and closure confirmation. Closure criteria shall be defined in the Problem Management Strategy or relevant organizational procedures.

7.2.8.3.2.5 Communication and Reporting

The status of all open and closed problems shall be **monitored and reported** to stakeholders (e.g., project management, QA, acquirer) at regular intervals. Reports shall include summaries of critical problems, trends, and resolution timelines.

7.2.8.4 Deliverables

The following deliverables shall be produced during this process:

- Problem Management Strategy or Plan;
- Categorization and prioritization scheme;
- Problem Reports (PRs) with unique identifiers;

- Root cause analyses and impact assessments;
 - Problem trend analysis reports;
 - Problem resolution records and closure logs;
 - Status reports and dashboards for internal and external stakeholders;
 - Updated configuration and test artifacts linked to resolved problems.
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7.2.8.5 Audit and Configuration Control

All problem reports, analyses, resolutions, and related evidence shall be placed under **configuration management**.

Problem resolution records shall be retained for the duration required by organizational policy or regulatory obligations.

Audits shall periodically verify that problems are categorized correctly, resolved in compliance with procedures, and tracked through closure.

Digital signatures, access control, and traceability links shall be used to protect problem records against tampering or loss.

Cross-References

This process interacts with the following:

- **Software Requirements Analysis (7.1.2)** — for identifying and resolving requirement-related defects.
- **Software Construction (7.1.5), Integration (7.1.6), and Qualification Testing (7.1.7)** — for reporting, tracking, and resolving issues discovered during development and testing.
- **Software Quality Assurance (7.2.3)** — for tracking non-conformances and feeding results into QA records.
- **Software Verification (7.2.4) and Software Validation (7.2.5)** — for recording and resolving defects discovered during testing and operational evaluation.
- **Software Review (7.2.6) and Software Audit (7.2.7)** — for capturing and processing findings from reviews and audits.
- **Software Configuration Management (7.2.2)** — for controlling changes arising from problem resolutions and ensuring proper baselining.
- **Project Risk and Measurement Processes (6.3)** — for feeding trend data into project-level risk mitigation and continuous improvement.